Common Half Yearly Examination - December 2025

Standard 12

CHEMISTRY

Time: 3.00 Hours Marks: 70

Part - I

Choose the most appropriate answer: **15x1=15**

1) If 50% of a first order reaction was completed in 60 minutes, 75% of the same reaction under the same conditions would be completed in
   a) 90 min b) 30 min c) 60 min d) 120 min
2) Which one of the following is a biodegradable polymer?
   a) HDPE b) PVC c) PHBV d) Nylon - 6
3) The colour of UO_2^{2+} is
   a) red b) yellow c) green d) pink
4) A complex in which the oxidation number of the metal is zero, is
   a) $K_3[Fe(CN)_6]$ b) $K_4[Fe(CN)_6]$ c) a and b d) $Fe(CO)_5$
5) Which of the following can act as lowry bronsted acid as well as base
   a) HPO_4^{2-} b) HCl c) SO_4^{2-} d) Br^-
6) Which of the following electrolytic solution has the least specific conductance
   a) 2N b) 0.002 N c) 0.02 N d) 0.2 N
7) In froth floatation, sodium ethyl xanthate is used as a
   a) collector b) depressing agent c) flux d) frothing agent
8) The vacant space in bcc latice unit cell is
   a) 48% b) 23% c) 32% d) 26%
9) In which of the following reactions new carbon-carbon bond is not formed
   a) aldol condensation b) Friedal Craft reaction
   c) Kolbe's reaction d) Wolf Kishner reaction
10) Carbon atoms in fullerence with formula C_{60} have
   a) sp^3 hybridised b) sp hybridised c) sp^2 hybridised
   d) partially sp^2 and partially sp^3 hybridised
11) **Assertion (A) :** Among halogen acids HF has low melting and boiling points
   Reason (R) : In HF hydrogen bond is present
   a) Assertion is true but reason is wrong
   b) Assertion is wrong but reason is true
   c) Both assertion and reason are ture. Reason is correct explanation of Assertion
   d) both assertion and reason are false
12) Which one of the following is the strong acid
   a) 2 nitrophenol b) 4 chloro phenol c) 4 nitro phenol d) 3 nitro phenol
13) The product formed by the reaction of an aldehyde with a primary amine
   a) Schiff's base b) Carboxylic acid c) aromatic acid d) ketone
14) α-D(+) Glucose and β-(D)+ glucose are
   a) Anomers b) Epimers
   c) Enantiomers d) confirmational isomers
15) The change of W/o emulsion into O/w emulsion is called
   a) coagulation b) emulsification
   c) decomposition d) inversion of phase

Part - II

Answer any six questions. Question no. 24 is compulsory: **6x2=12**

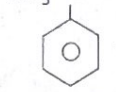
16) What is Leaching?
17) Write a short note on anomolous properties of the first element of p-block.
18) What are inter halogen compounds. Give an example.
19) Write the pH value of the following substances:
   a] vinegar b] black coffee c] baking soda d] soapy water

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20) Lan Fe^{3+} oxidises Bromide to Bromine under standard conditions?

Given $E^{\circ}\frac{Fe^{3+}}{Fe^{2+}} = 0.771\,V$, $E^{\circ}\frac{Br_2}{Br^-} = 1.09\,V$

21) Identify x and y 

22) What are Zuitter ions. Give an example.
23) What are the advantages of food additives.
24) The rate of formation of a dimer in a second order reaction is 7.5×10^{-3} mol $L^{-1}S^{-1}$ at 0.05 mol L^{-1} monomer concentration. Calculate the rate constant.

Part - III

Answer any six questions. Question no. 33 is compulsory: **6x3=18**

25) Give reason:
   i) Why Fluorine is more reactive than other halogens.
   ii) HF acid is not stored in glass bottles. Why?
26) What are the effects of lanthanide contraction?
27) Give any three characteristics of ionic crystals.
28) Explain Psuedo first order reaction with an example.
29) Explain Intermediate compound formation theory of catalysis with an example.
30) Identify A, B and C
   Ethanoic acid $\xrightarrow{SOCl_2}$ (A) $\xrightarrow[BaSO_4]{Pd}$ B $\xrightarrow{NaOH}$ C
31) Account for the following:
   i) Ethylamine is soluble in water whereas aniline is not.
   ii) PKb of aniline is more than that of methylamine.
32) Give any three differences between DNA and RNA.
33) Write (i) the oxidation state (ii) nature of ligand (iii) electronic configuraiton in octahedral crystal field for the complex $K_4\,[Mn(CN)_6]$

Part - IV

Answer all the questions: **5x5=25**

34) a] i) Write about calcination.
   ii) Explain the electrometallurgy of Aluminium.

(OR)

   b] Describe the structure of diborane.

35) a] i) Explain chromyl chloride test.
   ii) Out of $Lu(OH)_3$ and $La(OH)_3$ which is more basic and why?

(OR)

   b] i) $[Ti(H_2O)_6]^{3+}$ is coloured while $(Sc(H_2O)_6)^{3+}$ is colourless. Explain.
   ii) Give the difference between double salts and co-ordination compounds.

36) a] Outline the classification of the point defects. Explain any one type of the point defects.

(OR)

   b] i) Give the differences between order and molecularity of a reaction
   ii) Write the relation between Ionic product(s) and solubility product (ksp)?

37) a] i) State Faraday's Laws of electrolysis.
   ii) Why are lyophilic colloidal sols are more stable than lyophobic colloidal sol.

(OR)

   b] Explain Lucas test of differentiating three types of alcohols.

38) a] How will you prepare (i) malachite green from beuzaldehyde
   (ii) Benzoic acid from Toluene (iii) Aniline from Nitrobenzene

(OR)

   b] i) Write a short note on peptide bond.
   ii) How is Nylon 6 prepared? Mention its use.

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